Name: ____________________

STATISTICS, (HISTOGRAMS)

Date: ________________

LO: I can draw and interpret histograms with unequal class widths using frequency density.

Histograms

A **histogram** displays continuous data using bars where the **area** of each bar represents frequency. The y-axis shows **frequency density**, not frequency.

Formula: Frequency density = frequency ÷ class width. Frequency = frequency density × class width.

Quick examples

●	FD = frequency ÷ class width	frequency density formula
●	Frequency = FD × class width	reverse calculation
●	Area of bar = frequency	area represents count
●	Height of bar = frequency	height = frequency density
	All bars must be the same	

Key idea

Frequency density = frequency ÷ class width. The AREA of each bar (not height) represents the frequency.

FD = f ÷ w

f = frequency, w = class width

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – calculating frequency density

Class 10-20 has frequency 30. Find the frequency density.

Class width = 20 - 10 = 10

FD = frequency ÷ class width

FD = 30 ÷ 10 = 3

FD = 3

Divide frequency by class width.

Example 2 – finding frequency from histogram

A bar has height (FD) 4 and width 5. Find the frequency.

Frequency = FD × width

= 4 × 5

= 20

Frequency = 20

Multiply frequency density by class width to get frequency.

Example 3 – completing a table

Class 0-10, frequency 15; class 10-30, frequency 40. Find FDs.

0-10: width = 10, FD = $\frac{15}{10}$ = 1.5

10-30: width = 20, FD = $\frac{40}{20}$ = 2

Draw bars at heights 1.5 and 2

FD: 1.5 and 2

Different class widths give different frequency densities.

Example 4 – estimating from histogram

A histogram bar covers 20-35 with FD = 2.4. Estimate the number in 20-25.

Total frequency for 20-35: 2.4 × 15 = 36

Assume uniform: proportion in 20-25 = $\frac{5}{15}$

Estimate = 36 × $\frac{5}{15}$ = 12

Approximately 12

Assume data is evenly spread within each class for estimates.

YOUR TURN – PRACTICE (deliberate practice)

1. Calculating frequency density

Find the frequency density for each class.

a) Class 0-5, frequency 20

b) Class 5-15, frequency 30

c) Class 15-25, frequency 45

d) Class 25-50, frequency 50

2. Finding frequency

Find the frequency from the frequency density and class width.

a) FD = 3, class width = 10

b) FD = 1.5, class width = 20

c) FD = 5, class width = 4

d) FD = 2.5, class width = 8

3. Completing tables

Complete the frequency density column.

a) 0-10 (f=25), 10-20 (f=35), 20-40 (f=30)

b) 0-5 (f=10), 5-15 (f=40), 15-30 (f=45)

c) 0-20 (f=60), 20-30 (f=50), 30-50 (f=80)

d) 0-4 (f=12), 4-10 (f=24), 10-25 (f=30)

4. Reading histograms

A histogram has these bars. Find the total frequency.

a) 0-10 FD=2, 10-30 FD=3, 30-50 FD=1

b) 0-5 FD=4, 5-20 FD=2, 20-25 FD=6

c) 0-15 FD=3, 15-25 FD=5, 25-40 FD=2

d) 0-8 FD=2.5, 8-12 FD=4, 12-20 FD=1.5

5. Estimation problems

Use the histogram data to estimate.

a) Bar 10-30, FD=3. Estimate frequency for 10-15.

b) Bar 0-20, FD=2. Estimate frequency for 5-10.

c) Bar 20-50, FD=4. Estimate frequency for 30-40.

d) Bar 0-12, FD=5. Total frequency in 0-12?

e) Two bars: 0-10 FD=3, 10-25 FD=2. Total?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) Frequency density = frequency ÷ class width ☐

Correct answer: __________

Reason: __________________

b) The height of a bar in a histogram represents the frequency ☐

Correct answer: __________

Reason:

c) The area of a bar represents the frequency ☐

Correct answer: __________

Reason: __________________

d) All classes in a histogram must be the same width ☐

Correct answer: __________

Reason: __________________

e) Frequency = frequency density × class width ☐

Correct answer: __________

Reason: __________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A histogram shows journey times. Class 0-10 has FD 1.5, class 10-20 has FD 3, class 20-40 has FD 2, class 40-60 has FD 0.5. Find the total number of journeys and estimate the percentage that took less than 15 minutes.

Expression: ______________________________

Simplified: ______________________________

b) 80 students took a test. The results are shown in a histogram with classes: 0-30 (FD = 0.4), 30-50 (FD = 1.5), 50-70 (FD = k), 70-100 (FD = 0.6). Find k.

Expression: ______________________________

Simplified: ______________________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!